

Task No.: 10**Date: 08-02-2026**

Title: Implementation of Accelerometer and Proximity Sensors**Aim:**

To develop an Android Application that implements Accelerometer and proximity sensors.

Code:*MainActivity.kt*

```
package com.example.madtask10
```

```
import android.content.Context
import android.hardware.Sensor
import android.hardware.SensorEvent
import android.hardware.SensorEventListener
import android.hardware.SensorManager
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import kotlin.math.abs
```

```
class MainActivity : AppCompatActivity(), SensorEventListener {
```

```
    lateinit var sensorManager: SensorManager
```

```
    var accelerometer: Sensor? = null
```

```
    var proximitySensor: Sensor? = null
```

```
    lateinit var btnAcc: Button
```

```
    lateinit var btnProx: Button
```

```
    lateinit var btnClear: Button
```

```
    lateinit var txtData: TextView
```

```
var currentMode = ""

override fun onCreate(savedInstanceState: Bundle?) {

    super.onCreate(savedInstanceState)

    setContentView(R.layout.activity_main)

    btnAcc = findViewById(R.id.btnAcc)
    btnProx = findViewById(R.id.btnProx)
    btnClear = findViewById(R.id.btnClear)
    txtData = findViewById(R.id.txtData)

    sensorManager =
        getSystemService(Context.SENSOR_SERVICE) as SensorManager

    accelerometer =
        sensorManager.getDefaultSensor(Sensor.TYPE_ACCELEROMETER)

    proximitySensor =
        sensorManager.getDefaultSensor(Sensor.TYPE_PROXIMITY)

    btnAcc.setOnClickListener {

        if (accelerometer == null) {

            Toast.makeText(this,
                "Accelerometer not available",
                Toast.LENGTH_SHORT).show()

        } else {

            currentMode = "ACC"

            sensorManager.unregisterListener(this)

            sensorManager.registerListener(
                this,
                accelerometer,
                SensorManager.SENSOR_DELAY_NORMAL
            )
        }
    }
}
```

```

    }
}

```

```

btnProx.setOnClickListener {

```

```

    if (proximitySensor == null) {

```

```

        Toast.makeText(this,
            "Proximity Sensor not available",
            Toast.LENGTH_SHORT).show()

```

```

    } else {

```

```

        currentMode = "PROX"

```

```

        sensorManager.unregisterListener(this)

```

```

        sensorManager.registerListener(
            this,
            proximitySensor,
            SensorManager.SENSOR_DELAY_NORMAL
        )
    }
}

```

```

btnClear.setOnClickListener {

```

```

    sensorManager.unregisterListener(this)

```

```

    txtData.text = "Sensor Data cleared"

```

```

    currentMode = ""

```

```

}

```

```

override fun onSensorChanged(event: SensorEvent?) {

```

```

    if (event == null) return

```

```

    if (currentMode == "ACC"

```

```

        && event.sensor.type == Sensor.TYPE_ACCELEROMETER) {

```

```

val x = event.values[0]
val y = event.values[1]
val z = event.values[2]

val status =
    if (abs(x) > 1 || abs(y) > 1 || abs(z - 9.8f) > 1)
        "Device Moving"
    else
        "Device Stable"

```

```

txtData.text =
    "Accelerometer:\n\n" +
        "X: $x\n" +
        "Y: $y\n" +
        "Z: $z\n\n" +
        "Status: $status"

```

```

}

```

```

if (currentMode == "PROX"
    && event.sensor.type == Sensor.TYPE_PROXIMITY) {

```

```

    val distance = event.values[0]

```

```

    val message =
        if (distance < proximitySensor!!.maximumRange)
            "Object Near\nDistance: $distance cm"
        else
            "Object Away\nDistance: $distance cm"

```

```

    txtData.text = message

```

```

}

```

```

}

```

```

override fun onAccuracyChanged(sensor: Sensor?, accuracy: Int) {}

```

```

override fun onPause() {
    super.onPause()
    sensorManager.unregisterListener(this)
}

```

```

}

```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center_horizontal"
        android:orientation="vertical">

        <TextView
            android:id="@+id/txtTitle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Name: Yoga Shri P K"
            android:textSize="22sp"
            android:textStyle="bold"
            android:gravity="center"/>

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Roll No.: 23BCS192"
            android:textSize="20sp"
            android:textStyle="bold"
            android:gravity="center"/>

    </LinearLayout>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:gravity="center"
        android:orientation="vertical">
```

```
<Button
    android:id="@+id/btnAcc"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:text="ACCELEROMETER"/>
```

```
<Space
    android:layout_width="match_parent"
    android:layout_height="20dp"/>
```

```
<Button
    android:id="@+id/btnProx"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:text="PROXIMITY SENSOR"/>
```

```
<Space
    android:layout_width="match_parent"
    android:layout_height="30dp"/>
```

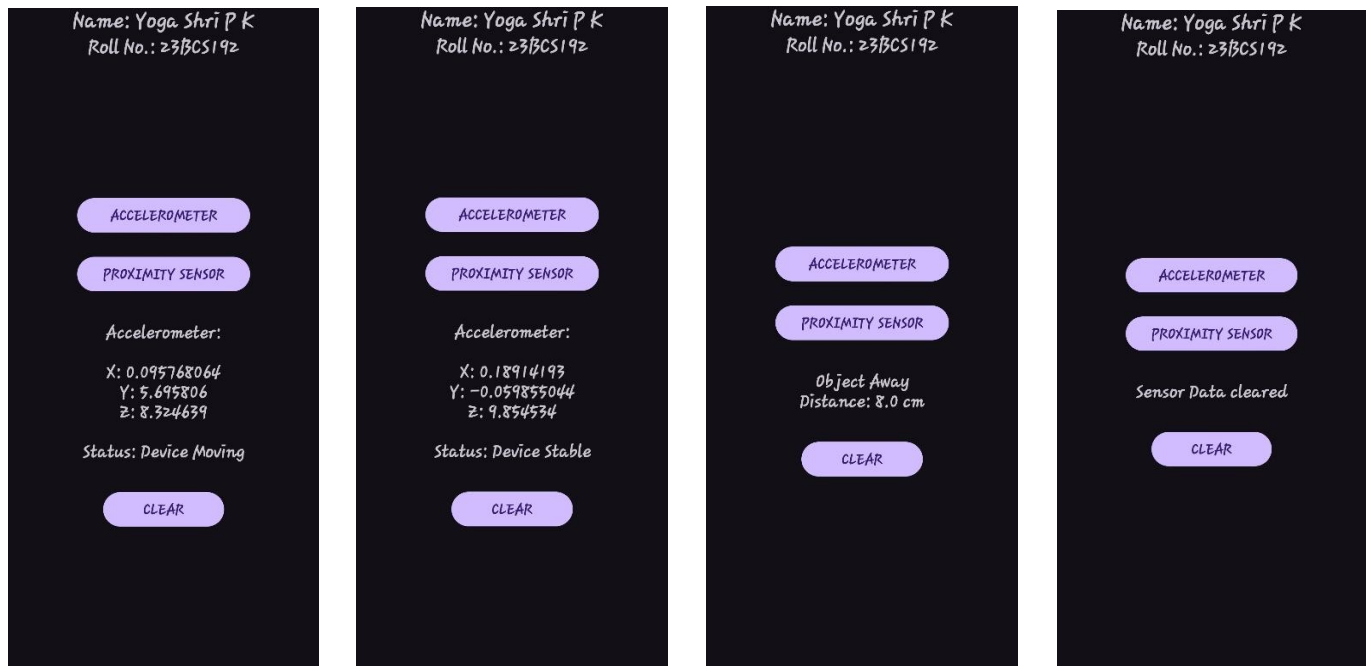
```
<TextView
    android:id="@+id/txtData"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Sensor Data will appear here"
    android:textSize="18sp"
    android:gravity="center"/>
```

```
<Space
    android:layout_width="match_parent"
    android:layout_height="30dp"/>
```

```
<Button
    android:id="@+id/btnClear"
    android:layout_width="140dp"
    android:layout_height="wrap_content"
    android:text="CLEAR"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

Output:**Result:**

Thus, Android Application that implements accelerometer and proximity sensor is developed and executed successfully.